



Management of patient with Blood disorders

Dr . Bardes Shaban



الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

Hematology

OUTLINE

- ◉ Definition of hematology
- ◉ Functions of blood
- ◉ Components of blood
- ◉ Characteristics of blood
- ◉ Techniques to control bleeding

OUTLINE / **BLOOD DISORDERS**

◎ Anemia

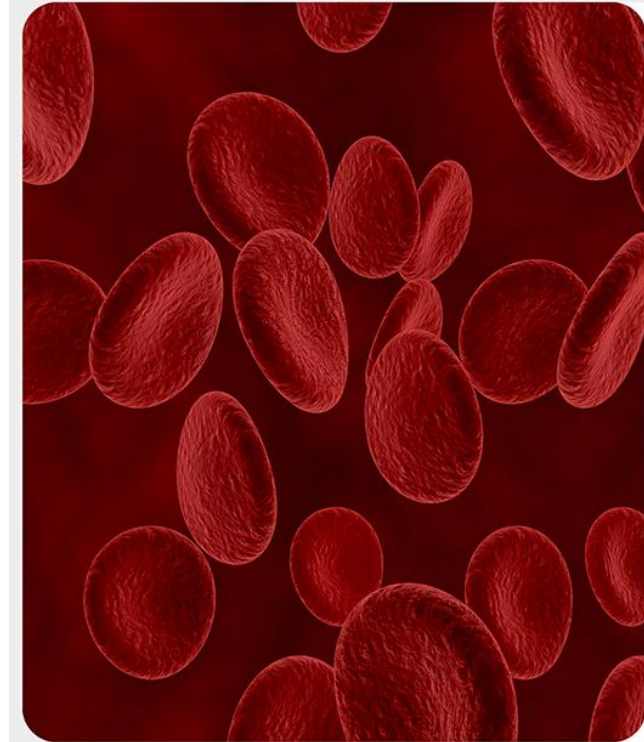
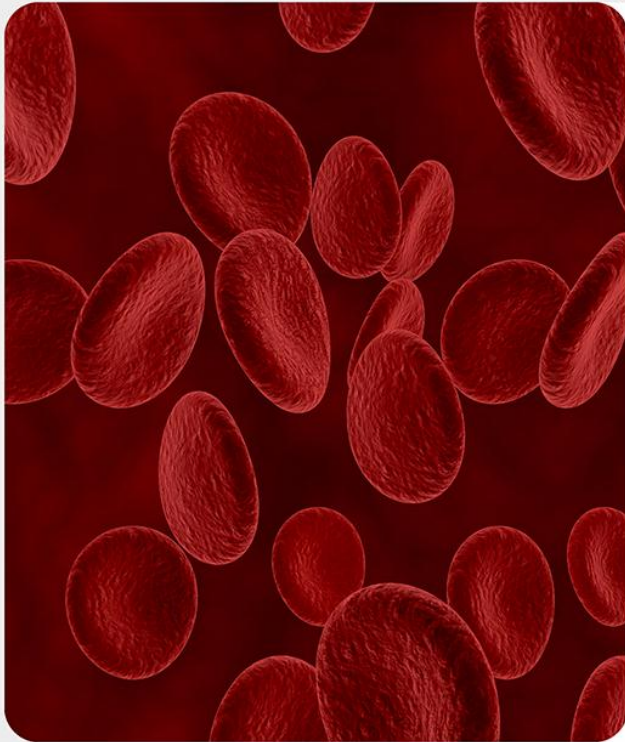
1. Iron deficiency anemia
2. Pernicious anemia
3. Folic acid deficiency
4. Aplastic anemia
5. Sickle cell anemia

◎ Leukemia

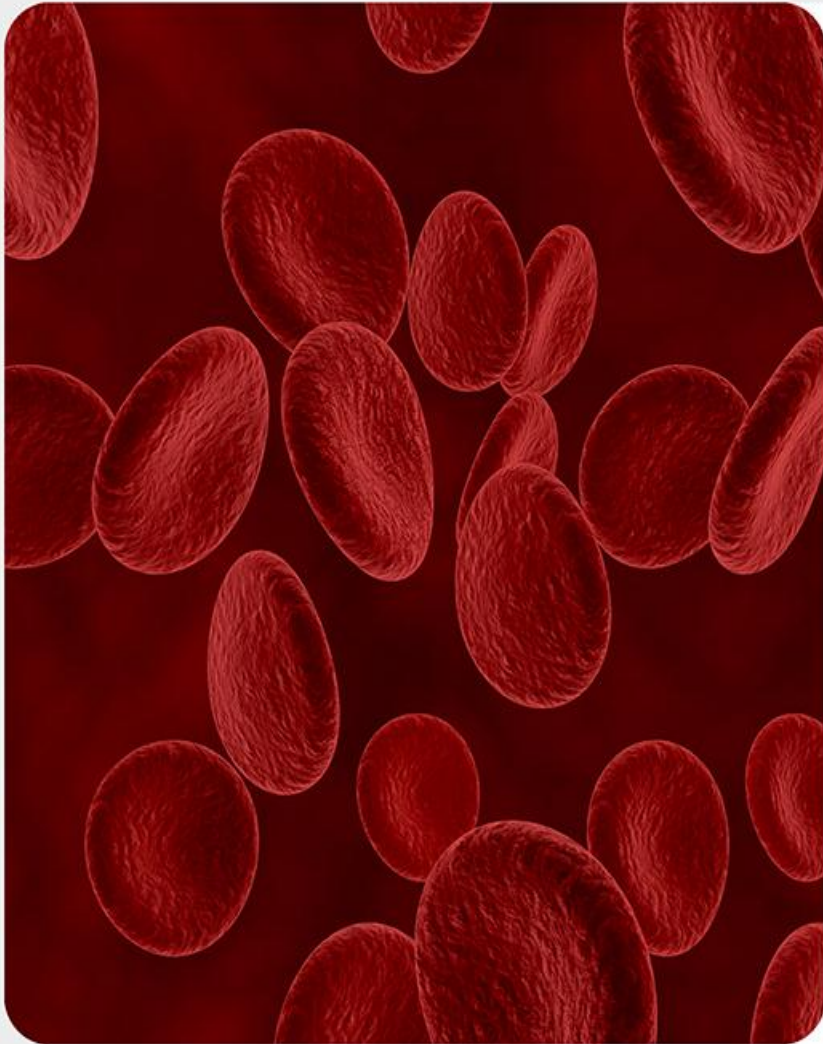
◎ Thrombocytopenia

Hematology

It is the Study of blood, blood components, and blood-forming tissues



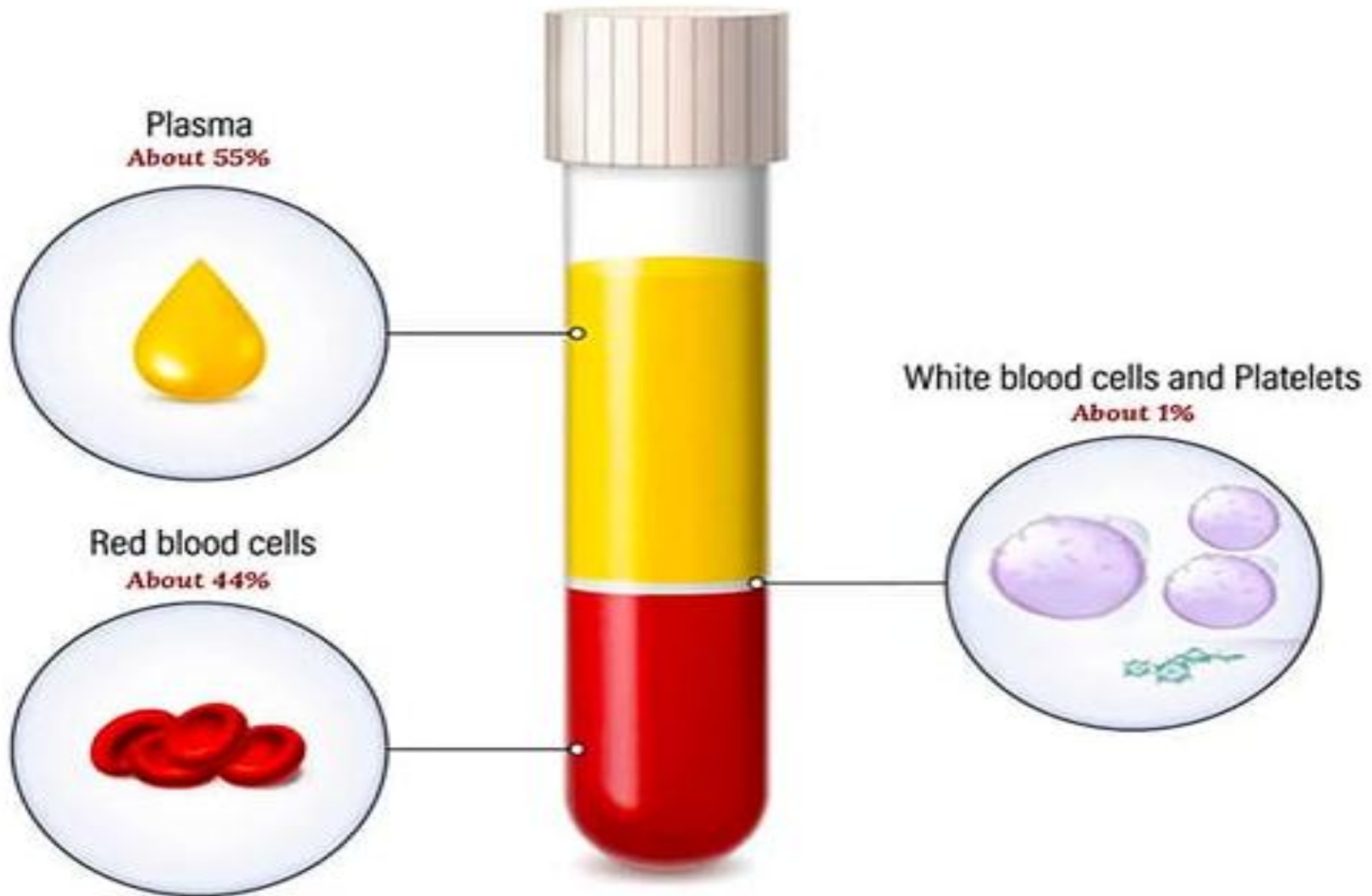
FUNCTIONS OF THE **BLOOD**



Blood has several functions in the body, including **transportation** of **oxygen** and **nutrients** to tissues, removal of waste products, regulation of body temperature, and protection against infections and injuries through the immune system.

Blood also helps maintain homeostasis by regulating pH levels and electrolyte balance in the body.

Component of blood



Blood Cells



Platelets



Thrombocytes

Red Blood Cells



Erythrocytes

White Blood Cells



Basophil



Neutrophil



Eosinophil



Monocyte



Lymphocytes

Characteristics of blood

- **Volume:** Female: 4-5 L _ Male: 5-6 L
- **Temperature:** 38 °C (100.4 F)
- **PH:** 7.35- 7.45
- **Viscosity:** Whole blood: 4.5- 5.5_ plasma : 2
- **Specific gravity:** 1.048 : 1.066

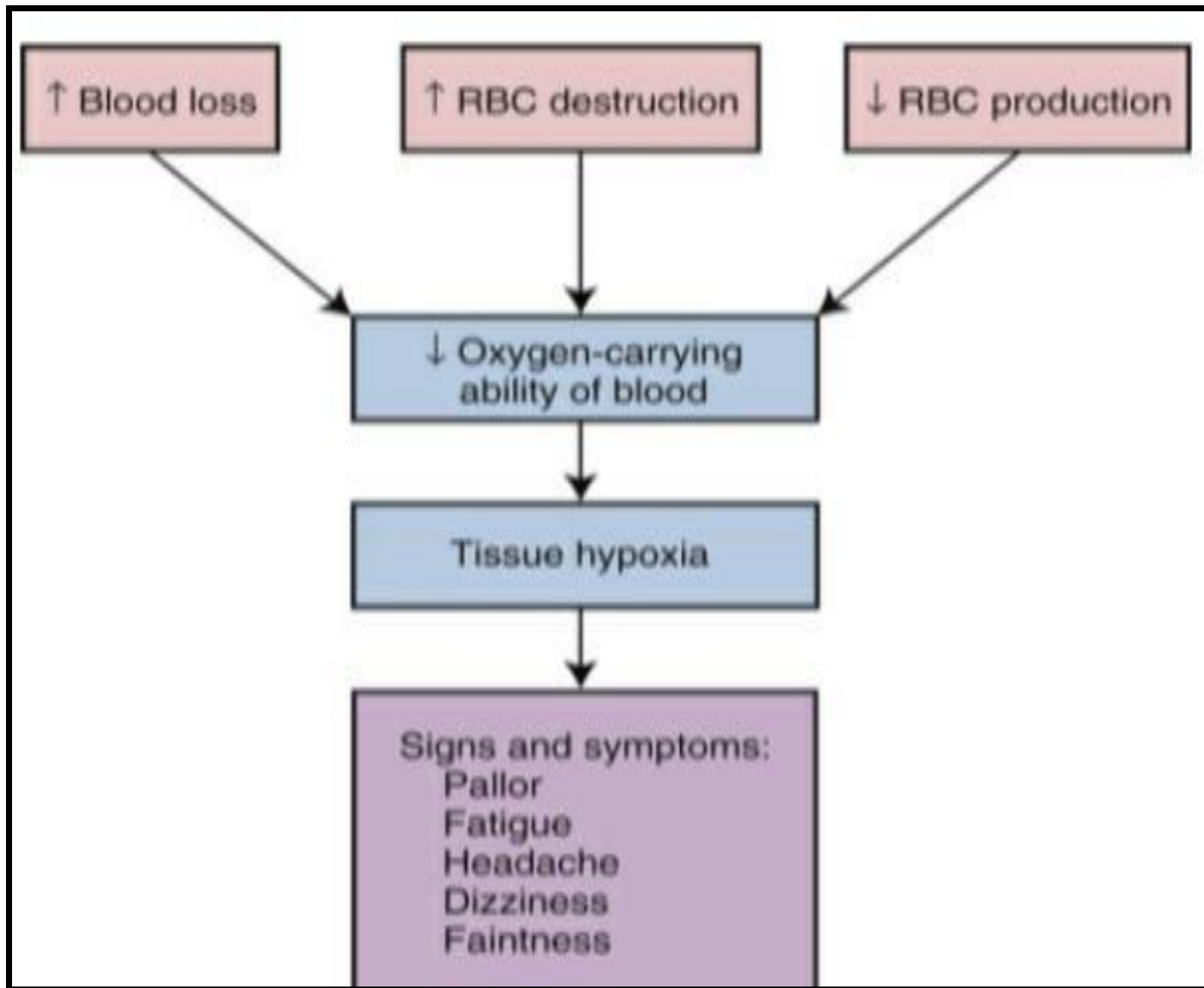
Techniques to control bleeding



Pressure with Part elevation

Anemia

- ◉ Anemia is a state in which there are insufficient numbers and functioning RBCs, or a lack of hemoglobin, to meet the demands of the tissues for oxygen.
- ◉ May be related to :
 - * Blood loss
 - * Failure in blood cell production
 - * Excessive destruction of red cells (hemolysis)



1- Iron deficiency anemia

Insufficient **iron** to produce hemoglobin

○ Predisposing factors:

- Acute or chronic **blood loss**
- **Insufficient** iron diet/ **unhealthy** dieting
- Iron **mal absorption**
- Need for **exceed** iron as in **growth** and **pregnancy**.



Fatigue



Shortness of breath



Weakness

Iron Deficiency Symptoms



**Heart
palpitations**



Poor focus



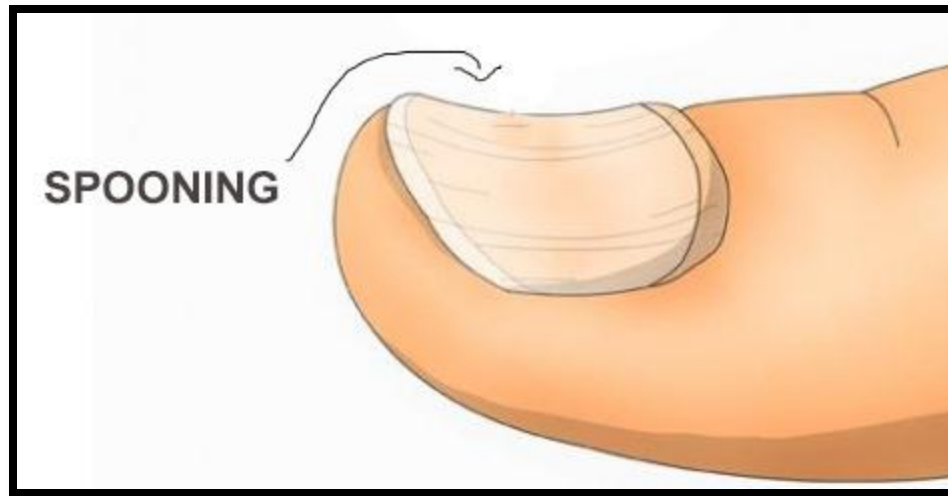
**Restless leg
syndrome**



Headaches

Signs & Symptoms :

- ◉ Brittleness of hair, spoon shaped nails



- ◉ Decreased : CBC , hemoglobin, hematocrit, and serum iron levels.

Treatment

❖ Medical / Iron supplements

Oral iron

- Monitor for (anorexia, nausea ,vomiting , abdominal pain, diarrhea or constipation, melena)
- Avoid taking iron with an antacid as it interferes with iron absorption
- Administer with orange juice or vitamin C to facilitate absorption.

Parenteral iron

- Avoid massaging of injection site (IM)

Monitor for (pain at injection site, localized abscess, fever and chill.

Blood transfusion

in acute, severe anemia



Nursing Intervention

- Monitor for **signs of bleeding** of all body systems
- Encourage **increased iron diet** (green leafy vegetables, organ meat, legumes,)
- **Avoid** tea and coffee because it **impairs** iron absorption.
- Assist the patient with self-care **activities**
- Teach the patient about **adverse reactions** to oral iron therapy, such as **black stools**, **constipation** or **diarrhea**, and GI **disturbances**

2- Pernicious anemia

Anemia resulting from **vitamin B12** deficiency.

Predisposing factors

- ◉ Hereditary factors
- ◉ Autoimmune
- ◉ Crohn's disease
- ◉ Subtotal / total gastrectomy

Crohn's disease



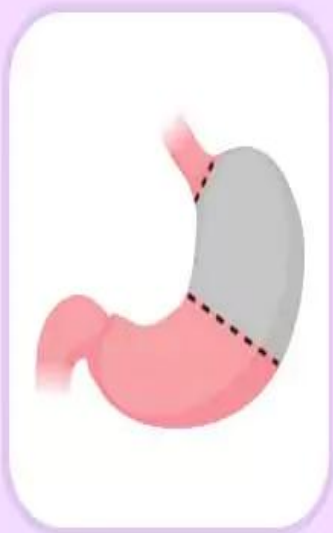
Gastrectomy



Total
Gastrectomy



Subtotal
Gastrectomy



Proximal
Gastrectomy



Distal
Gastrectomy



Laparoscopic
Gastrectomy

Signs & Symptoms :

- Numbness & paresthesia of the hands, feet weakness because vitamin B12 is necessary for normal neurological function
- red sore in tongue.



Treatment

❖ Medical

Vitamin B12 supplementation

(Oral, sublingual, IM and deep SC)

Folic acid supplements.

- To produce new RBCs and to cover vitamin B12 deficiency.

Nursing Intervention

- Ensure that the patient is protected from **burns** and trauma **because** of a diminished sensation to **heat** and **pain** because of neurologic impairment.
- Careful follow-up of neurologic impairment as in **diabetic** patients.
- **Plus interventions of Iron deficiency anemia.**

3- Folic Acid Deficiency Anemia

- Folic acid is required for (DNA) synthesis leading to the formation and maturation of RBCs

Causes of folic acid deficiency

- Dietary deficiency, lack of leafy green vegetables and citrus fruits
- Mal absorption
- Drugs that interfere with absorption as (anti-seizure)
- hemodialysis treatments, because folic acid is lost during dialysis

Treatment

Medical

- Folic acid supplements

Nursing

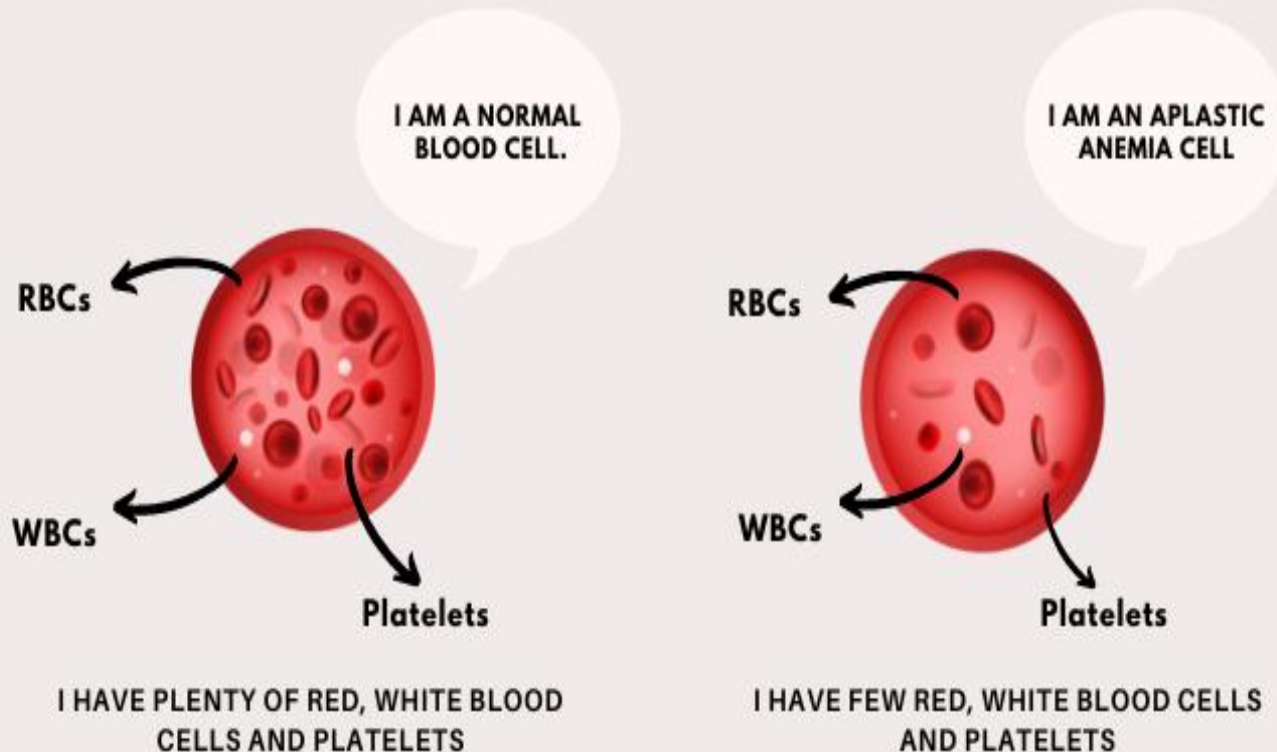
- Encourage the patient to eat foods containing folic acid as leafy green vegetables, citrus fruits, and legumes
- Inform patient that folic acid may change urine color to dark yellow.
- Plus interventions of vit.B12 deficiency anemia.

4- Aplastic Anemia

◉ Definition: Stem cell disorder that lead to bone marrow depression, inadequate production of the RBCs, platelets, and WBCs.

Aplastic anemia

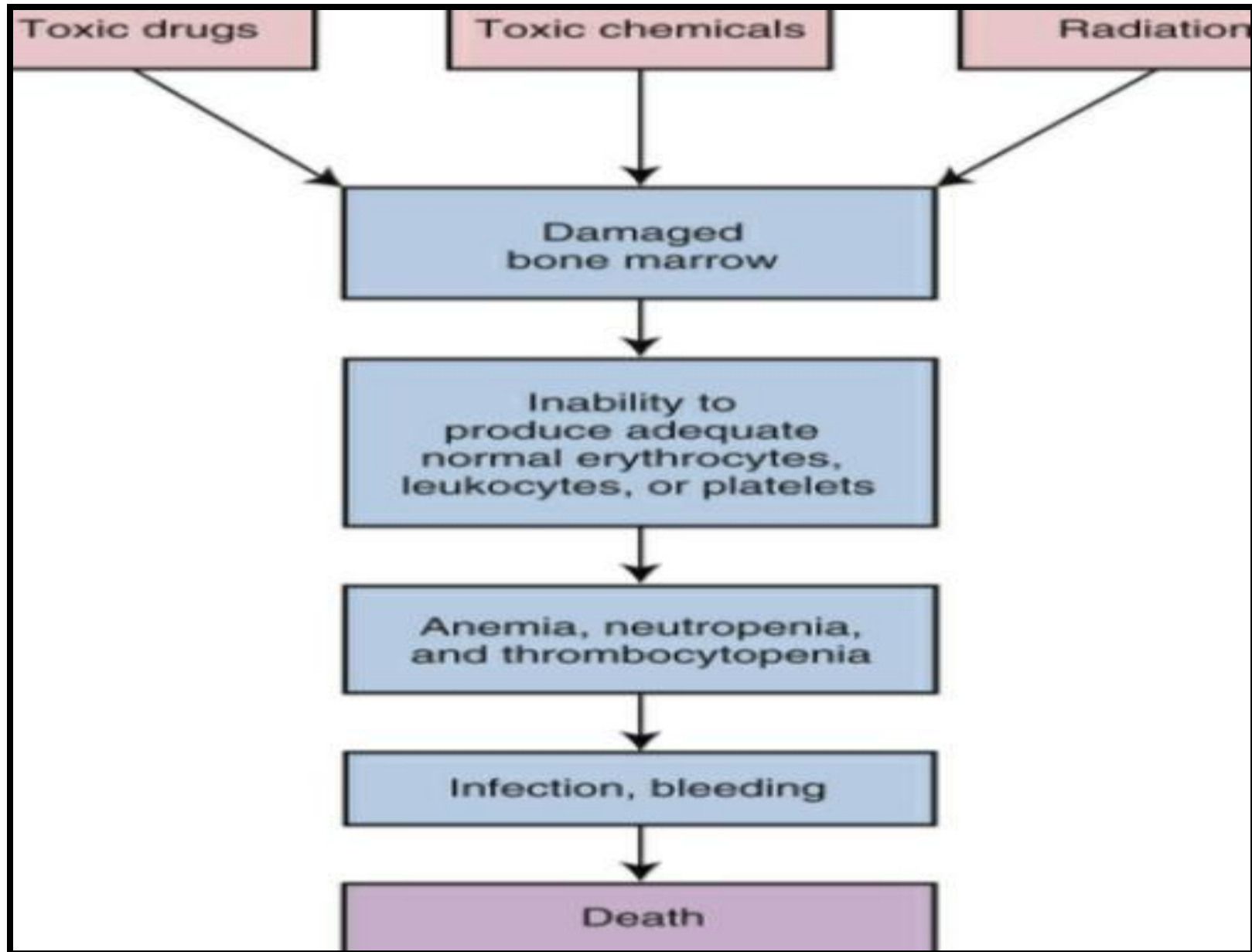
A DISEASE IN WHICH THE BONE MARROW DOES NOT PRODUCE ENOUGH BLOOD CELLS.



Predisposing factors:

- Autoimmune basis
- Congenital chromosomal alterations
- Chemo-Radio therapy
- Exposure to toxic substances as benzene and insecticides.

PATHOPHYSIOLOGY OF APLASTIC ANEMIA



**SYMPTOMS
OF APLASTIC
ANEMIA**

**Skin
Rash**



Headache



Fatigue



**Irregular
Heartbeats**



**Uncontrolled
Bleeding
from Minor cuts**



Dizziness



Medical ,Surgical &Nursing Management

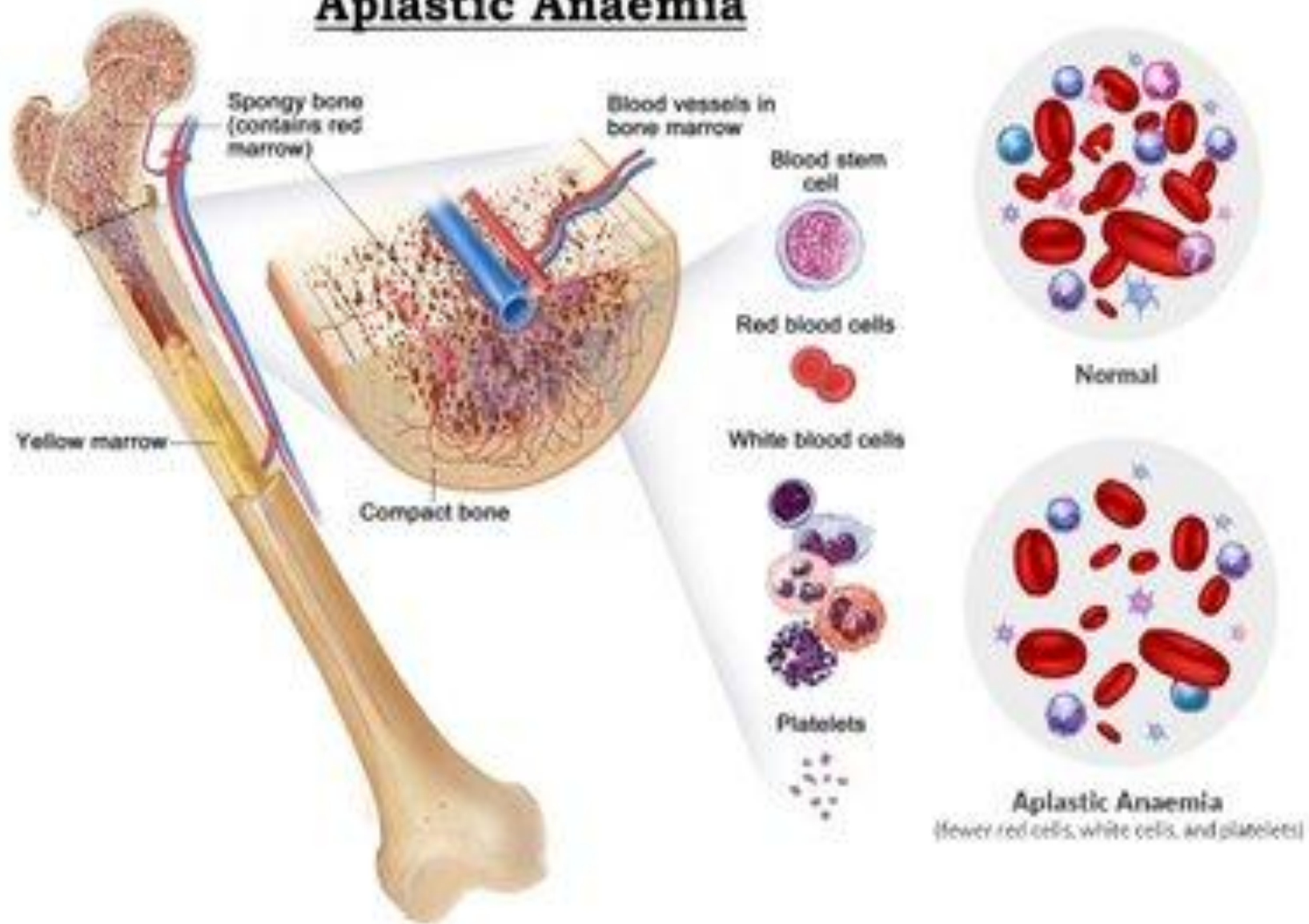
Medical

- Steroids
- Erythropoietin
- Neupogen
- Immunosuppressive therapy

Surgical

- Bone marrow transplantation
- Hematopoietic stem cell transplant (HSCT)

Aplastic Anaemia



Nursing Intervention

Maintain bleeding precautions

- Use a soft toothbrush, Avoid flossing.
- Avoid invasive procedures e.g. enemas,
- Avoid intramuscular injections.
- Maintain pressure on intravenous (IV) for 5 minutes.
- Encourage use of shoes when out of bed.

- Avoid use of drugs that interfere with platelet function, (aspirin & NSAID)
- Administer stool softeners as ordered to prevent straining.
- Move and turn patient gently.
- Instruct patients to avoid blowing the nose
- Avoid use of Foley catheters and other invasive devices.

Protect Pt. from infection

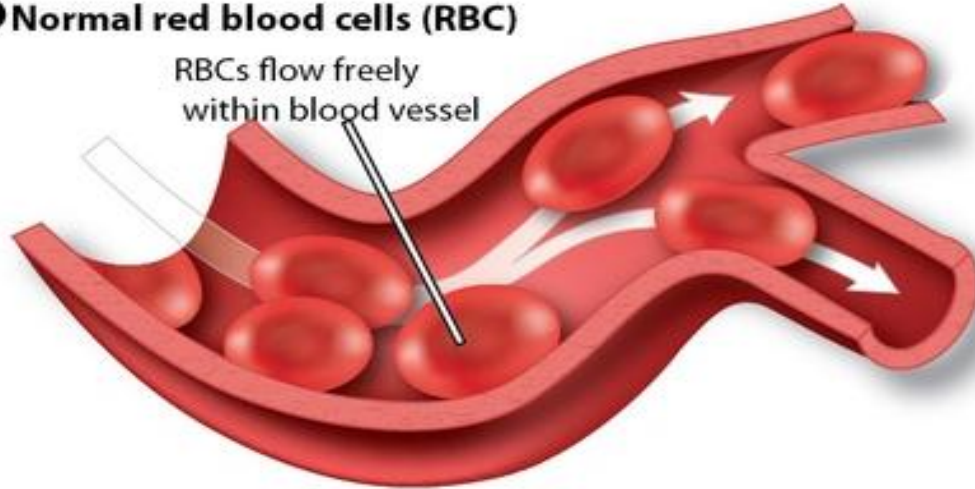
- Place the patient in a private room & assure that all personnel , visitors wash hands before entering the room. Also, Pt wash hands before & after using the toilet, before & after eating.
- Prevent staff or visitors with known infections from entering the patient's room.
- Not handle flowers or plants brought into the room

5- Sickle cell Anemia/ hemolytic anemia

- An **inherited** anemia in which Erythrocytes become sickle- or crescent-shaped.
- Normal red cells live about **120** days.
- Sickled cells are thin wall & easily broken, so survive only about **15 to 20** days.

Sickle cell anemia

● Normal red blood cells (RBC)

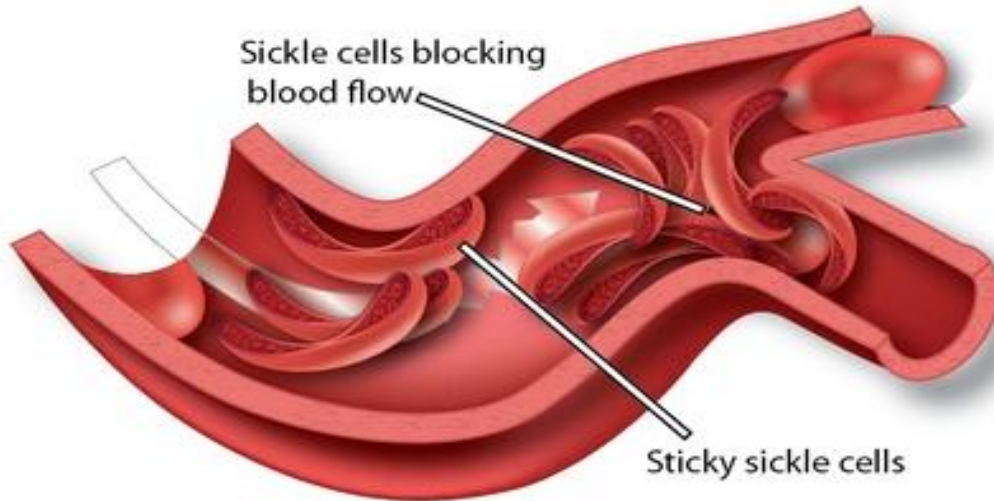


Cross-section of RBC



Normal hemoglobin

● Abnormal, sickled, red blood cells (sickle cells)



Cross-section of sickle cell



Abnormal hemoglobin form strands that cause sickle shape



Clinical manifestation & complications

- **Acute chest syndrome;** pneumonia ,coughing, wheezing, tachypnea, and chest pain.
- **Hand & foot syndrome** :an unequal growth of fingers and toes.



- When **more** and **more** , **sudden** , and **severe** sickling occurs, the blood becomes sluggish and does not flow easily, called **sickle cell crisis** .

Common symptoms of sickle cell crises :

- **Severe pain and swelling** in the joints, especially of the **elbows and knees**
- **Abdominal pain** with swelling of the spleen
- **Hypoxia**

Medical ,Surgical &Nursing Management

Medical

- Oral penicillin
- Blood transfusions
- Oxygen therapy
- Vasodilator agent

Surgical

- Bone marrow transplantation
- Splenectomy.

Nursing Intervention

- Administer **O2**, medication & blood transfusion as recommended.
- **Avoid tight** clothing that could impair circulation.
- Provide **bed rest** to decrease O2 use.
- Encourage patient to eat foods **high in calories**, with **folic acid** supplementation.
- Keep arms and legs from extreme cold.
- **Decrease** emotional stress.

Leukemia

- It is **malignant** disorders of blood characterized by the **uncontrolled accumulation** of **dysfunctional WBCs**
- Classifications of leukemia ; according to:
- **Type of white blood cell** that is multiplying:
lymphocytes , **granulocytes** , or **monocytes**.
- How **quickly it progresses**: **Acute** or **Chronic**

Causes of leukemia



Unknown
Hereditary

Immunosuppression

**Exposure to
chemotherapy
or radiotherapy**

Clinical Manifestations

- Severe infections
- Enlarged spleen, liver, and lymph nodes
- Weakness
- Renal pain
- Headache

Nursing Priorities of care

- Prevent infection & injury
- Maintain circulating blood volume.
- Alleviate pain.
- Promote optimal physical functioning.
- Provide psychological support.
- Provide information about disease processes, prognosis, and treatment needs.
- Conserve the client 's energy.

Thrombocytopenia

It is a condition characterized by an abnormally **low number of platelets** in the blood.

- Normal platelet count: 150,000–450,000/ μL
- Thrombocytopenia: $<150,000/\mu\text{L}$
- Severe if $<50,000/\mu\text{L}$

Causes

**Decreased
platelet
production**

**Increased
platelet
destruction**

**Platelet
sequestration**

Infections

Medications

Clinical manifestations

Skin bleeding

**Mucosal
bleeding**

**Prolonged
bleeding after
injury**

**Internal
bleeding**

Complications:

- Severe bleeding (internal and external)
- Intracranial hemorrhage
- Gastrointestinal bleeding
- Anemia due to blood loss
- Fatal hemorrhage if untreated

Medical & Nursing Management

Medical

- Avoid NSAIDs
- Avoid anticoagulants
- Blood transfusions

Nursing Intervention

- **Monitor for signs of bleeding:**

- Petechiae, ecchymosis, gum/nose bleeding
- Blood in urine or stool



- **Minimize trauma:**

- Use soft toothbrush, electric razor
- Avoid invasive procedures if possible

Nursing Intervention

- **Educate patients:**
 - Avoid contact sports
 - Report any unusual bleeding
- **Monitor vitals and lab values regularly**

Bleeding Disorders

Bleeding disorders are conditions that affect the way the blood clots, often due to defects in platelets, clotting factors, or blood vessel integrity.

Causes

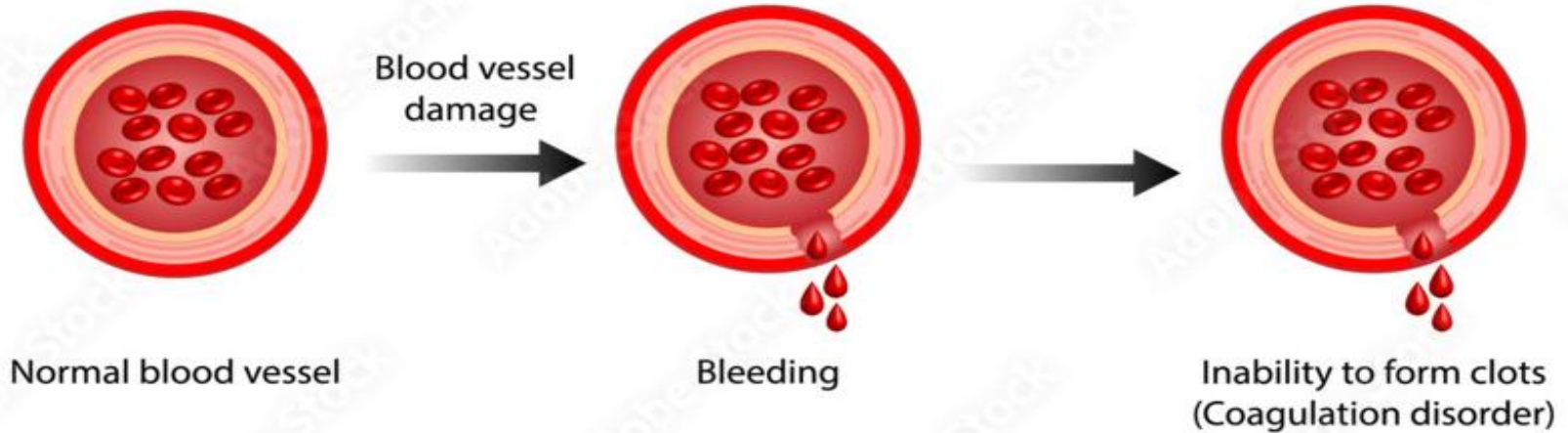
Inherited

Hemophilia A

Hemophilia B

vonWillebranddisease

Hemophilia



Types of Hemophilia

Hemophilia A



The most common type, affects 1 in every 5,000 births of people assigned male at birth.

Hemophilia B

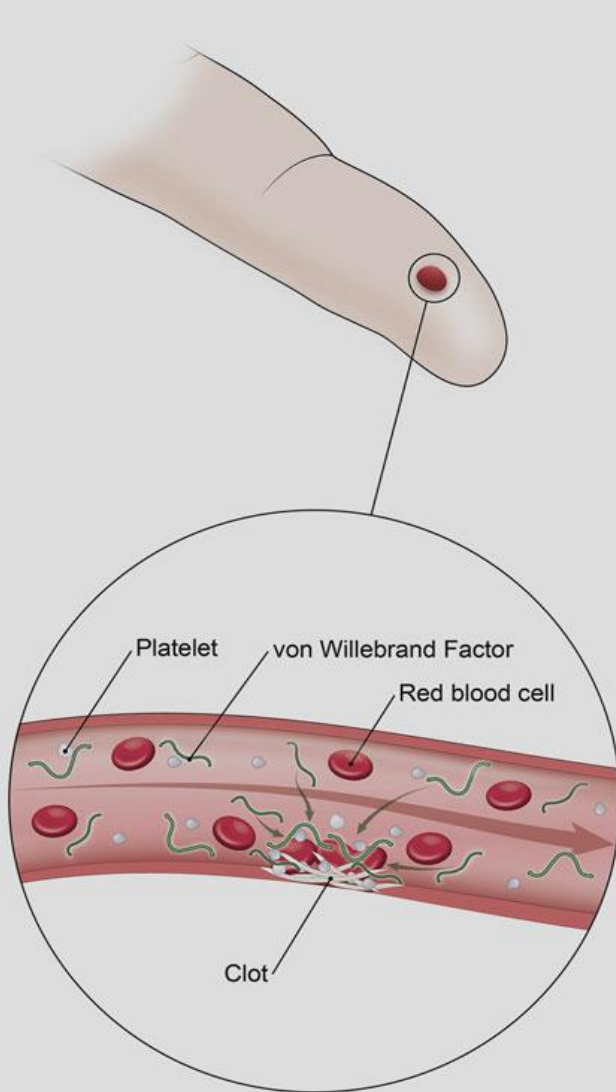


Also called as Christmas disease, it is four times less common than Hemophilia A.

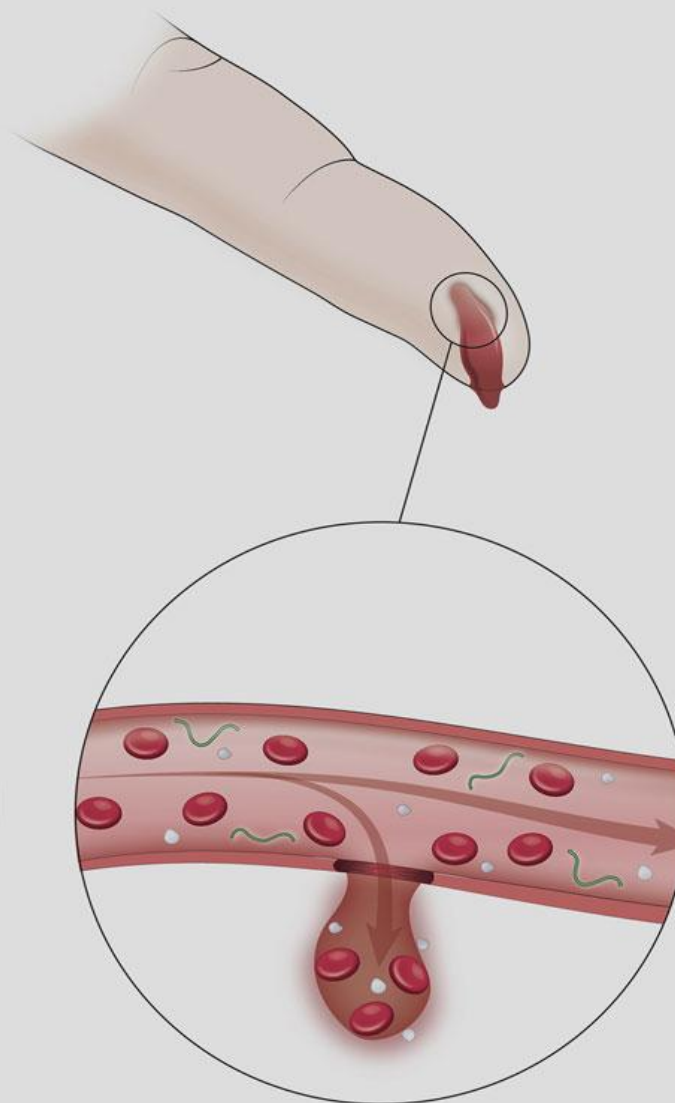
Hemophilia C



Also called Rosenthal syndrome, it affects 1 in every 100,000 people.



Normal



von Willebrand Disease

Symptoms of Hemophilia A



Nosebleeds



Bleeding gums
or mouth



Swollen and
stiff joints



Bruises



Coughing or
vomiting blood



Blood in
stool or urine



Heavy menstrual
bleeding

Causes

Acquired

Liver disease

Vitamin K deficiency

**Disseminated
Intravascular
Coagulation**

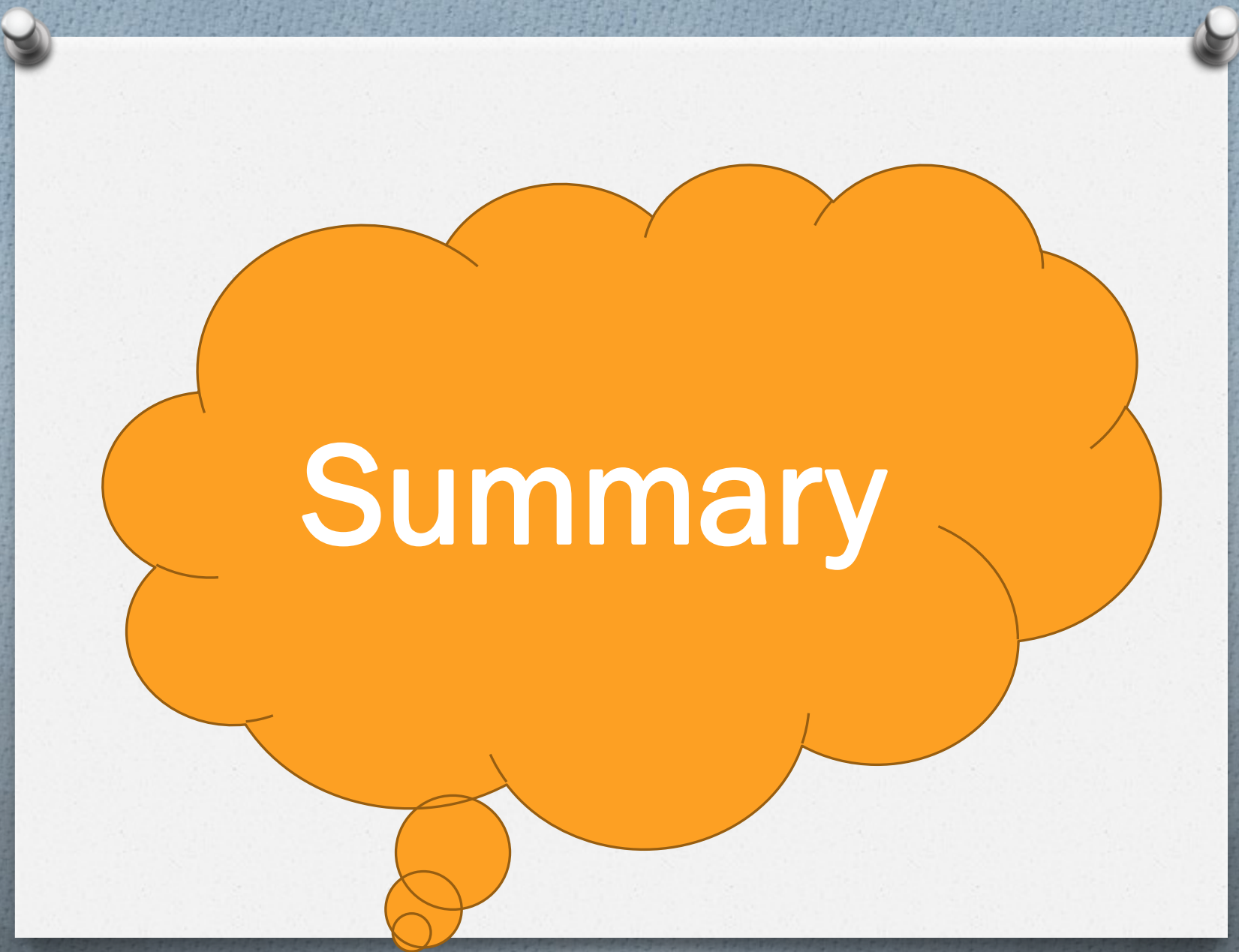
**Use of
anticoagulants**

Signs & Symptoms:

- bleeding ,Petechiae, epistaxis
- Hemarthrosis
- Deep tissue hematomas
- Delayed bleeding after trauma or surgery
- Heavy menstrual bleeding
- Prolonged bleeding post-dental
- Easy bruising

Nursing Care:

- Assess for and prevent bleeding
- Use pressure after injections or IVs
- Apply cold packs to bleeding sites
- Education about avoiding trauma
- Encourage use of medical alert bracelets
- Collaborating with hematology specialists



Summary



Choose the correct answer

Which type of anemia caused due to Stem cell disorder ?

A. Iron deficiency anemia

B. Pernicious anemia

C. Aplastic anemia

D. Sickle cell anemia

Which type of anemia caused due to vit B12 deficiency ?

A. Iron deficiency anemia

B. Pernicious anemia

C. Aplastic anemia

D. Sickle cell anemia

Complete

what are the manifestations of Iron deficiency anemia?

- Fatigue
- Palpitation
- weakness
- Headache
- Brittle hair
- Spoon like nail
- Poor focus

Complete

what are the side effects of oral Iron supplements ?

- Anorexia
- Nausea, vomiting
- Abdominal pain,
- Diarrhea or constipation,
- Melena
- Black stool

Thank
you